



NSW Education Standards Authority

2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6–16)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the value of π^{10} to two significant figures?

- A. 9.36×10^4
- B. 9.4×10^4
- C. 9.36×10^5
- D. 9.4×10^5

2 What values of x satisfy $4 - 3x \leq 12$?

- A. $x \leq -\frac{16}{3}$
- B. $x \geq -\frac{16}{3}$
- C. $x \leq -\frac{8}{3}$
- D. $x \geq -\frac{8}{3}$

3 What is the value of p so that $\frac{a^2 a^{-3}}{\sqrt{a}} = a^p$?

- A. -3
- B. $-\frac{3}{2}$
- C. $-\frac{1}{2}$
- D. 12

- 4 A parabola has vertex (2, 1) and focus (5, 1).

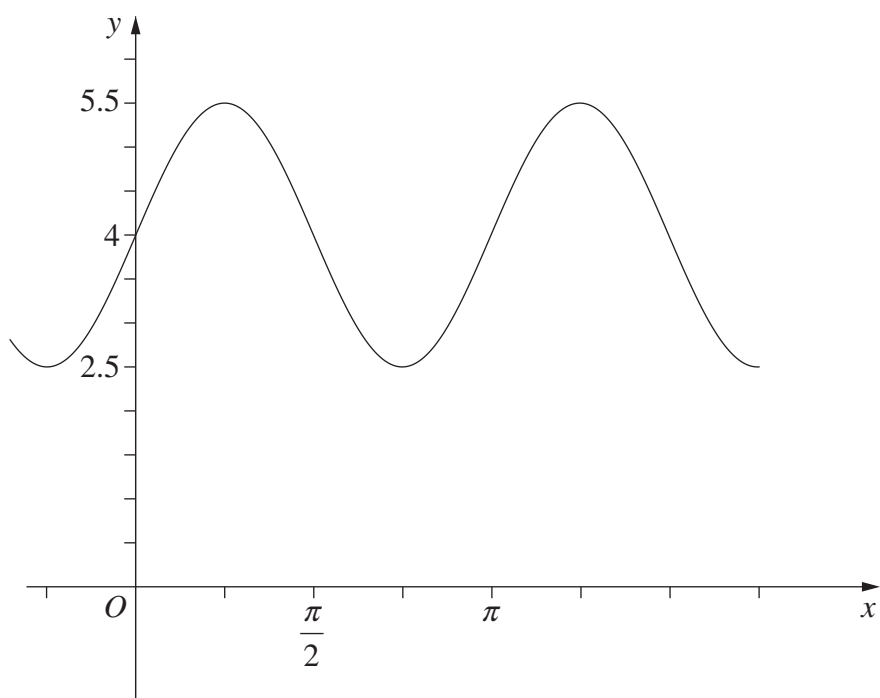
What is the equation of this parabola?

- A. $(x - 1)^2 = 12(y - 2)$
B. $(x - 2)^2 = 12(y - 1)$
C. $(y - 1)^2 = 12(x - 2)$
D. $(y - 2)^2 = 12(x - 1)$
- 5 Which of the following is equal to $\frac{\log_2 9}{\log_2 3}$?
- A. 2
B. 3
C. $\log_2 3$
D. $\log_2 6$
- 6 A game is played by tossing an ordinary 6-sided die and an ordinary coin at the same time. The game is won if the uppermost face of the die shows an even number or the uppermost face of the coin shows a tail (or both).

What is the probability of winning this game?

- A. $\frac{1}{4}$
B. $\frac{1}{2}$
C. $\frac{3}{4}$
D. 1

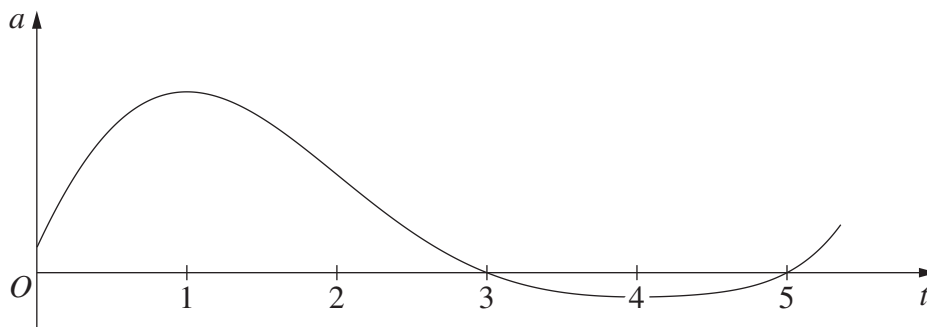
- 7 The diagram shows part of the graph of $y = a \sin(bx) + 4$.



What are the values of a and b ?

- A. $a = 3$ $b = \frac{1}{2}$
B. $a = 3$ $b = 2$
C. $a = 1.5$ $b = \frac{1}{2}$
D. $a = 1.5$ $b = 2$

- 8 A particle is moving along a straight line. The graph shows the acceleration of the particle.



For what value of t is the velocity v a maximum?

- A. 1
 - B. 2
 - C. 3
 - D. 5
- 9 Which expression is equal to $\int \tan^2 x \, dx$?
- A. $\tan x - x + C$
 - B. $\tan x - 1 + C$
 - C. $\frac{\tan^3 x^2}{6} + C$
 - D. $\frac{\tan^3 x}{3} + C$

- 10 A particle is moving along a straight line with displacement x at time t .

The particle is stationary when $t = 11$ and when $t = 13$.

Which of the following **MUST** be true in this case?

- A. The particle changes direction at some time between $t = 11$ and $t = 13$.
- B. The displacement function of the particle has a stationary point at some time between $t = 11$ and $t = 13$.
- C. The acceleration of the particle is 0 at some time between $t = 11$ and $t = 13$.
- D. The acceleration function of the particle has a stationary point at some time between $t = 11$ and $t = 13$.

Section II

90 marks

Attempt Questions 11–16

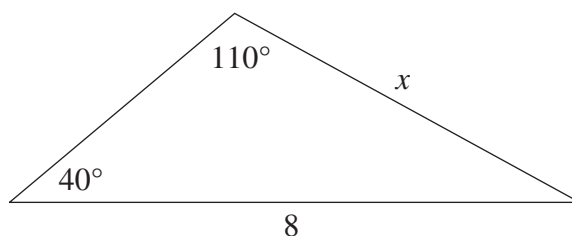
Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

- (a) Using the sine rule, find the value of x correct to one decimal place. **2**



- (b) Differentiate $x^2 \sin x$. **2**

- (c) Differentiate $\frac{2x + 1}{x + 5}$. **2**

- (d) What is the limiting sum of the following geometric series? **2**

$$2000 - 1200 + 720 - 432 \dots$$

- (e) Evaluate $\int_0^1 \frac{1}{(3x + 2)^2} dx$. **3**

Question 11 continues on page 7

Question 11 (continued)

- (f) A bag contains 5 green beads and 7 purple beads. Two beads are selected at random, without replacement. **2**

What is the probability that the two beads are the same colour?

- (g) The parabola $y = x^2$ meets the line $y = x + 2$ at the points $(-1, 1)$ and $(2, 4)$. **2**
Do NOT prove this.

By first sketching the graphs of $y = x^2$ and $y = x + 2$, shade the region which simultaneously satisfies the two inequalities $y \geq x^2$ and $y \geq x + 2$.

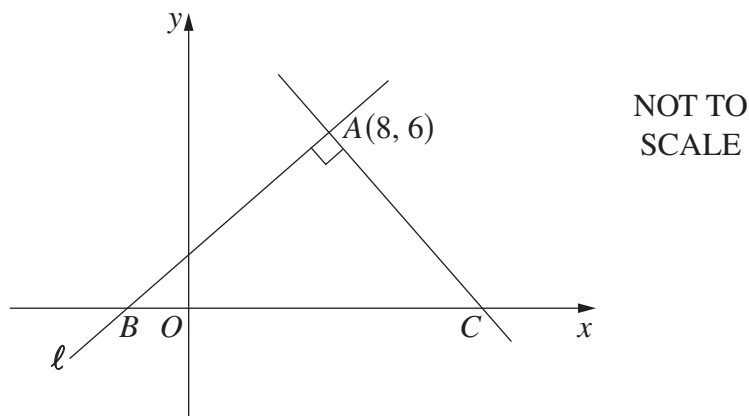
End of Question 11

Please turn over

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) The line ℓ , with equation $x - 2y + 4 = 0$, passes through the point $A(8, 6)$ and intersects the x -axis at the point B .

The line through A perpendicular to ℓ intersects the x -axis at the point C .



- (i) Find the equation of the line AC . 2
- (ii) Find the area of triangle ABC . 2
- (b) In an arithmetic series, the fourth term is 6 and the sum of the first 16 terms is 120. 3

Find the common difference.

- (c) The number of leaves, $L(t)$, on a tree t days after the start of autumn can be modelled by

$$L(t) = 200\,000e^{-0.14t}.$$

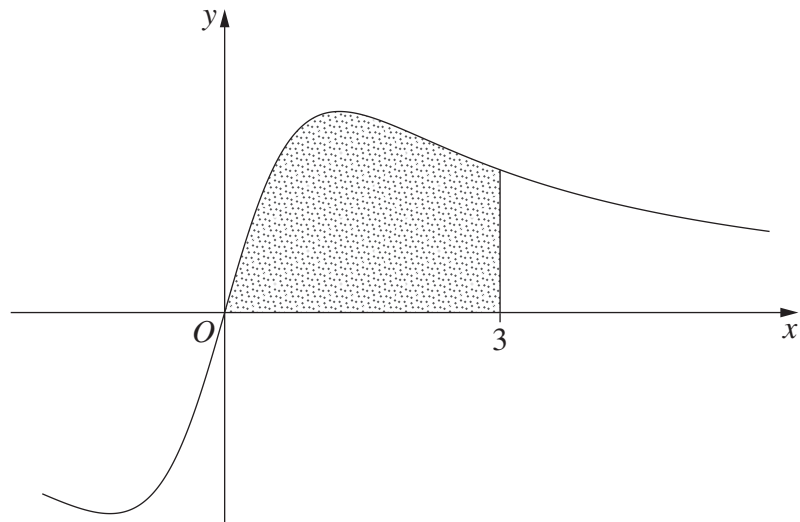
- (i) What is the number of leaves on the tree when $t = 31$? 1
- (ii) What is the rate of change of the number of leaves on the tree when $t = 31$? 2
- (iii) For what value of t are there 100 leaves on the tree? 2

Question 12 continues on page 9

Question 12 (continued)

- (d) The diagram shows the graph of $y = \frac{3x}{x^2 + 1}$.

3



The region enclosed by the graph, the x -axis and the line $x = 3$ is shaded.

Calculate the exact value of the area of the shaded region.

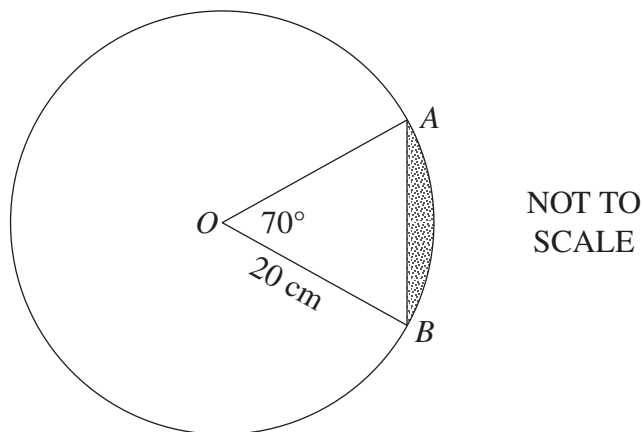
End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

(a) Solve $2 \sin x \cos x = \sin x$ for $0 \leq x \leq 2\pi$. **3**

(b) The diagram shows a circle with centre O and radius 20 cm. **3**

The points A and B lie on the circle such that $\angle AOB = 70^\circ$.



Find the perimeter of the shaded segment, giving your answer correct to one decimal place.

(c) (i) Differentiate $(\ln x)^2$. **2**

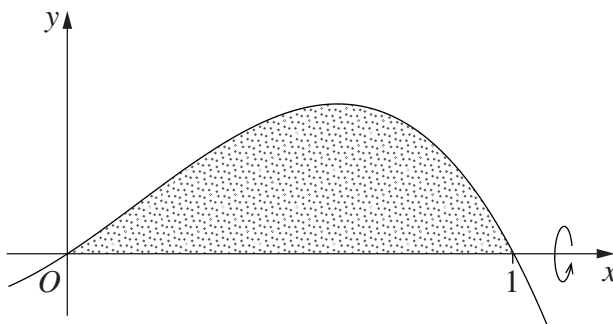
(ii) Hence, or otherwise, find **1**

$$\int \frac{\ln x}{x} dx.$$

Question 13 continues on page 11

Question 13 (continued)

- (d) The diagram shows the region bounded by the curve $y = x - x^3$, and the x -axis between $x = 0$ and $x = 1$. The region is rotated about the x -axis to form a solid. **3**



Find the exact value of the volume of the solid formed.

- (e) (i) Sketch the graph of $y = |x - 1|$ for $-4 \leq x \leq 4$. **1**
- (ii) Using the sketch from part (i), or otherwise, solve $|x - 1| = 2x + 4$. **2**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) A particle is moving along a straight line. The particle is initially at rest. The acceleration of the particle at time t seconds is given by $a = e^{2t} - 4$, where $t \geq 0$. 2

Find an expression, in terms of t , for the velocity of the particle.

- (b) The derivative of a function $y = f(x)$ is given by $f'(x) = 3x^2 + 2x - 1$.
- (i) Find the x -values of the two stationary points of $y = f(x)$, and determine the nature of the stationary points. 2

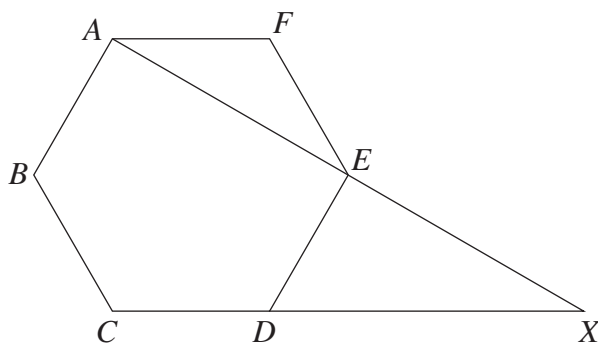
- (ii) The curve passes through the point $(0, 4)$. 2

Find an expression for $f(x)$.

- (iii) Hence sketch the curve, clearly indicating the stationary points. 2

- (iv) For what values of x is the curve concave down? 1

- (c) The regular hexagon $ABCDEF$ has sides of length 1. The diagonal AE and the side CD are produced to meet at the point X . 3



Copy or trace the diagram into your writing booklet.

Find the exact length of the line segment EX , justifying your answer.

- (d) The equation of the tangent to the curve $y = x^3 + ax^2 + bx + 4$ at the point where $x = 2$ is $y = x - 4$. 3

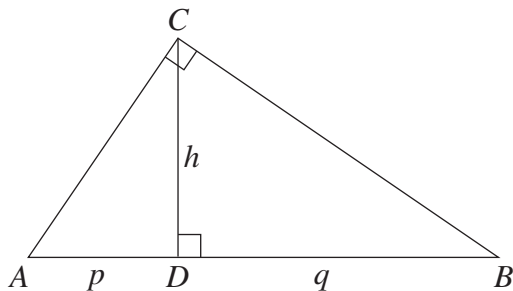
Find the values of a and b .

Question 15 (15 marks) Use the Question 15 Writing Booklet.

(a) Solve $e^{2\ln x} = x + 6$. **2**

- (b) The triangle ABC is a right-angled triangle with the right angle at C . The point D is chosen on AB so that CD is perpendicular to AB . **3**

The length of AD is p , the length of BD is q and the length of CD is h .



Show that $h = \sqrt{pq}$.

Question 15 continues on page 14

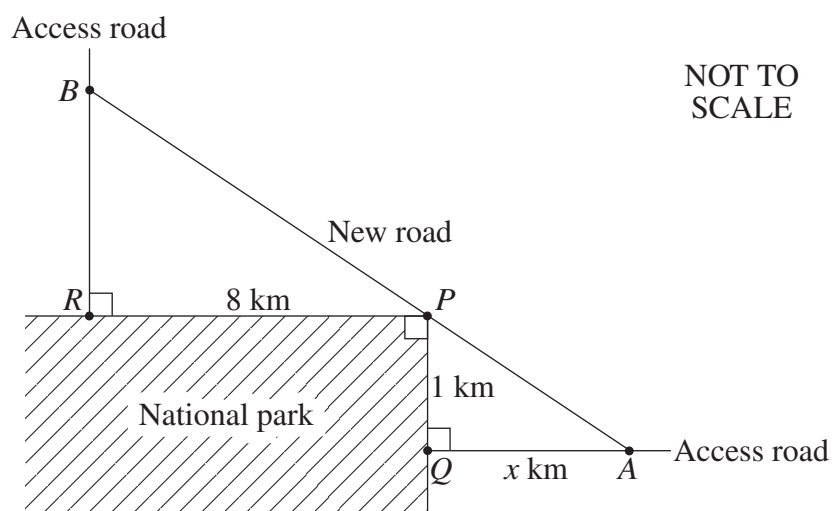
Question 15 (continued)

- (c) The entry points, R and Q , to a national park can be reached via two straight access roads. The access roads meet the national park boundaries at right angles. The corner, P , of the national park is 8 km from R and 1 km from Q . The boundaries of the national park form a right angle at P .

A new straight road is to be built joining these roads and passing through P .

Points A and B on the access roads are to be chosen to minimise the distance, D km, from A to B along the new road.

Let the distance QA be x km.



- (i) Show that $D^2 = (x + 8)^2 + \left(\frac{8}{x} + 1\right)^2$. 3
- (ii) Show that $x = 2$ gives the minimum value of D^2 . 3
- (d) The probability that a person chosen at random has red hair is 0.02.
- (i) Two people are chosen at random. 2
- What is the probability that at least ONE has red hair?
- (ii) What is the smallest number of people that can be chosen at random so that the probability that at least ONE has red hair is greater than 0.4? 2

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) A person wins \$1 000 000 in a competition and decides to invest this money in an account that earns interest at 6% per annum compounded quarterly. The person decides to withdraw \$80 000 from this account at the end of every fourth quarter. Let A_n be the amount remaining in the account after the n th withdrawal.

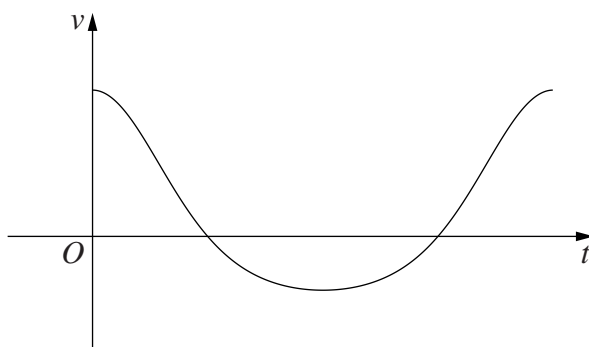
- (i) Show that the amount remaining in the account after the withdrawal at the end of the eighth quarter is **2**

$$A_2 = 1\,000\,000 \times 1.015^8 - 80\,000(1 + 1.015^4).$$

- (ii) For how many years can the full amount of \$80 000 be withdrawn? **3**

- (b) A particle moves in a straight line, starting at the origin. Its velocity, $v \text{ m s}^{-1}$, is given by $v = e^{\cos t} - 1$, where t is in seconds. **3**

The diagram shows the graph of the velocity against time.

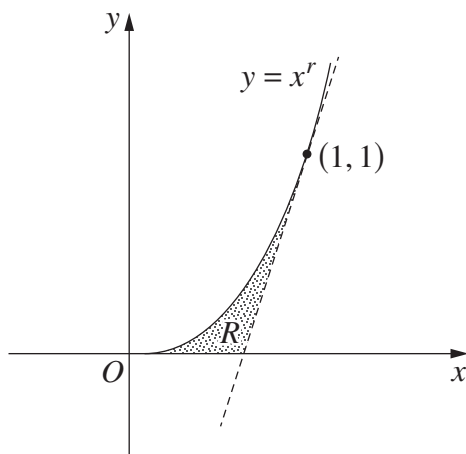


Using one application of Simpson's rule, estimate the position of the particle when it first comes to rest. Give your answer correct to two decimal places.

Question 16 continues on page 16

Question 16 (continued)

- (c) The diagram shows the region R , bounded by the curve $y = x^r$, where $r \geq 1$, the x -axis and the tangent to the curve at the point $(1, 1)$.



- (i) Show that the tangent to the curve at $(1, 1)$ meets the x -axis at $\left(\frac{r-1}{r}, 0\right)$. 2
- (ii) Using the result of part (i), or otherwise, show that the area of the region R is $\frac{r-1}{2r(r+1)}$. 2
- (iii) Find the exact value of r for which the area of R is a maximum. 3

End of paper



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2019 HIGHER SCHOOL CERTIFICATE EXAMINATION

REFERENCE SHEET

- Mathematics –
- Mathematics Extension 1 –
- Mathematics Extension 2 –

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

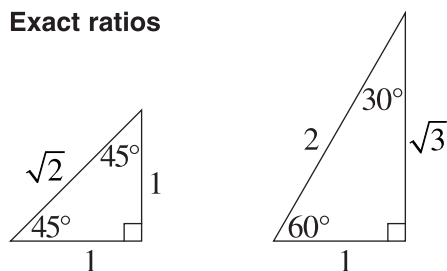
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \quad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

$$\text{tangent: } y = tx - at^2$$

$$\text{normal: } x + ty = at^3 + 2at$$

At (x_1, y_1) ,

$$\text{tangent: } xx_1 = 2a(y + y_1)$$

$$\text{normal: } y - y_1 = -\frac{2a}{x_1}(x - x_1)$$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

2019 HSC Mathematics Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	D
3	B
4	C
5	A
6	C
7	D
8	C
9	A
10	C

Section II

Question 11 (a)

Criteria	Marks
• Provides correct solution	2
• Correctly substitutes the given values into the sine rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{x}{\sin 40^\circ} &= \frac{8}{\sin 110^\circ} \\ x &= \frac{8 \sin 40^\circ}{\sin 110^\circ} \\ &= 5.47 \\ &= 5.5\end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the product rule, or equivalent merit	1

Sample answer:

$$\frac{d}{dx}(x^2 \sin x) = x^2 \cos x + 2x \sin x$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the quotient rule, or equivalent merit	1

Sample answer:

$$\frac{d}{dx}\left(\frac{2x+1}{x+5}\right) = \frac{(x+5)2 - (2x+1)}{(x+5)^2}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Identifies the common ratio, or equivalent merit	1

Sample answer:

$$2000 - 1200 + 720 - 432 \dots$$

$$a = 2000 \quad r = \frac{-1200}{2000}$$

$$= -0.6$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{2000}{1 - (-0.6)}$$

$$= 1250$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Obtains a primitive of the form $A(3x+2)^{-n}$, or equivalent merit	1

Sample answer:

$$\int_0^1 (3x+2)^{-2} dx$$

$$= \left[\frac{(3x+2)^{-1}}{-3} \right]_0^1$$

$$= \frac{-1}{3} \left[\frac{1}{3x+2} \right]_0^1$$

$$= \frac{-1}{3} \left(\frac{1}{5} - \frac{1}{2} \right)$$

$$= \frac{-1}{3} \left(\frac{-3}{10} \right)$$

$$= \frac{1}{10}$$

Question 11 (f)

Criteria	Marks
• Provides correct solution	2
• Correctly provides one relevant probability, or equivalent merit	1

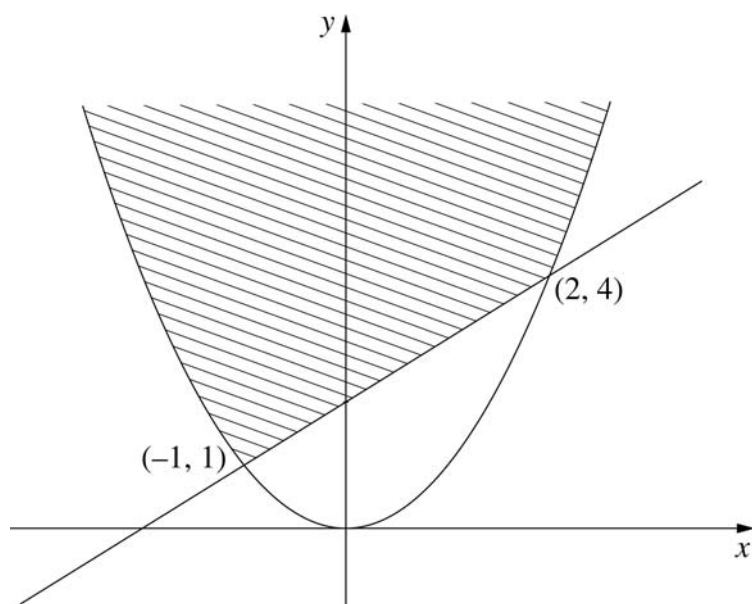
Sample answer:

$$\begin{aligned}
 &P(\text{gg}) \text{ or } P(\text{pp}) \\
 &= \frac{5}{12} \times \frac{4}{11} + \frac{7}{12} \times \frac{6}{11} \\
 &= \frac{20 + 42}{132} \\
 &= \frac{31}{66}
 \end{aligned}$$

Question 11 (g)

Criteria	Marks
• Provides correct solution	2
• Correctly indicates one region, or equivalent merit	1

Sample answer:



Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains the slope of the line AB , or equivalent merit	1

Sample answer:

$$2y = x + 4$$

$$y = \frac{1}{2}x + 2$$

Slope of line AC is $\frac{-1}{\frac{1}{2}} = -2$

$$y = -2x + K$$

Passes through $A(8, 6)$

$$6 = -2 \times 8 + K$$

$$K = 22$$

$$y = -2x + 22$$

$$(\text{or } 2x + y - 22 = 0)$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds the coordinates of B or C or the length of one side, or equivalent merit	1

Sample answer:

Height of $\triangle ABC$ is 6.

Base is distance between B and C .

$$B: \quad x - \text{int} \quad y = 0$$

$$x - 2 \times 0 + 4 = 0$$

$$x = -4$$

$$B(-4, 0)$$

$$C: \quad x - \text{int} \quad y = 0$$

$$2x + 0 - 22 = 0$$

$$x = 11$$

$$C(11, 0)$$

BC length is 15

$$\begin{aligned} \text{Area } \triangle ABC &= \frac{1}{2} \times 15 \times 6 \\ &= 45 \end{aligned}$$

Question 12 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains two equations involving a and d , or equivalent merit	2
• Obtains one equation involving a and d , or equivalent merit	1

Sample answer:

$$T_n = a + (n-1)d$$

$$6 = T_4 = a + 3d \quad \boxed{1}$$

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$120 = S_{16} = \frac{16}{2}(2a + 15d)$$

$$15 = 2a + 15d \quad \boxed{2}$$

$$6 = a + 3d \quad \boxed{1}$$

$$12 = 2a + 6d \quad 2 \times \boxed{1}$$

$$\boxed{2} - 2 \times \boxed{1} \quad 3 = 9d$$

$$d = \frac{1}{3}$$

Question 12 (c) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$L(31) = 200000e^{-0.14 \times 31}$$

$$= 2607.3\dots$$

$$= 2607 \text{ (nearest leaf)}$$

Question 12 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Correctly differentiates $L(t)$, or equivalent merit	1

Sample answer:

$$L'(t) = -0.14 \times 200\,000e^{-0.14t}$$

$$L'(31) = -0.14 \times 200\,000e^{-0.14 \times 31} (= -0.14L(31))$$

$$= -365.02\dots$$

Question 12 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains an exponential equation for t , or equivalent merit	1

Sample answer:

Want t so that $L(t) = 100$

$$100 = 200\,000e^{-0.14t}$$

$$\frac{1}{2000} = e^{-0.14t}$$

$$t = \frac{-1}{0.14} \ln\left(\frac{1}{2000}\right)$$

$$= 54.29\dots$$

100 leaves left when $t = 54.29\dots$

Question 12 (d)

Criteria	Marks
• Provides correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Obtains correct definite integral, or equivalent merit	1

Sample answer:

As function is above x -axis in given interval we have

$$\begin{aligned}
 \text{Area} &= \int_0^3 \frac{3x}{x^2 + 1} dx \\
 &= \frac{3}{2} \int_0^3 \frac{2x}{x^2 + 1} dx \\
 &= \frac{3}{2} [\ln(x^2 + 1)]_0^3 \\
 &= \frac{3}{2} (\ln(10) - \ln(1)) \\
 &= \frac{3}{2} \ln(10)
 \end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct solution	3
• Correctly solves one case, or equivalent merit	2
• Identifies both cases to be considered or finds one correct answer, or equivalent merit	1

Sample answer:

$$2 \sin x \cos x = \sin x$$

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x (2 \cos x - 1) = 0$$

$$\sin x = 0$$

$$2 \cos x - 1 = 0$$

$$x = 0, \pi, 2\pi$$

$$2 \cos x = 1$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

Question 13 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains the length of the chord AB , or equivalent merit	2
• Converts the angle at centre to radians or attempts to use the cosine rule, or equivalent merit	1

Sample answer:

$$AB^2 = 20^2 + 20^2 - 2 \times 20 \times 20 \cos 70$$

$$= 526.3838...$$

$$AB = 22.94$$

$$AB_{\text{Arc}} = 20 \times \frac{70\pi}{180}$$

$$= 24.43$$

$$P = 22.94 + 24.43$$

$$= 47.37$$

$$= 47.4 \text{ cm}$$

Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the chain rule, or equivalent merit	1

Sample answer:

$$\frac{d}{dx}(\ln x)^2 = 2 \ln x \times \frac{1}{x}$$

$$= \frac{2}{x} \ln x$$

Question 13 (c) (ii)

Criteria	Marks
Provides correct answer	1

Sample answer:

$$\begin{aligned}
 & \int \frac{\ln x}{x} dx \\
 &= \frac{1}{2} \int \frac{2}{x} \ln x dx \\
 &= \frac{1}{2} (\ln x)^2 + C
 \end{aligned}$$

Question 13 (d)

Criteria	Marks
Provides correct solution	3
Obtains correct primitive, or equivalent merit	2
Obtains correct integral, or equivalent merit	1

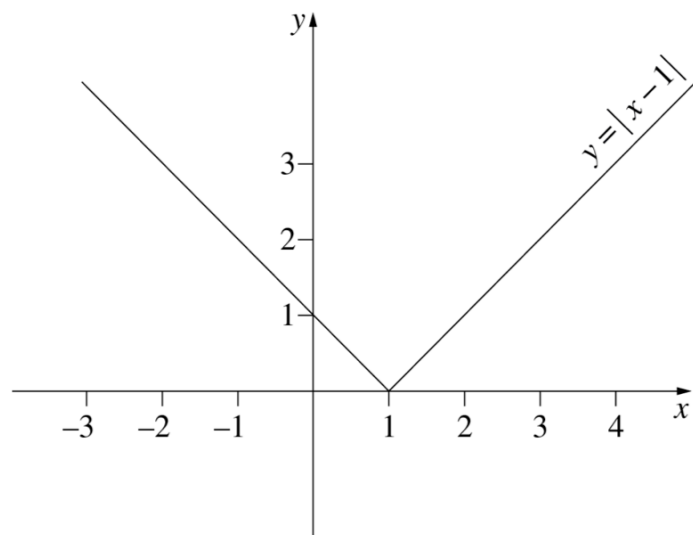
Sample answer:

$$\begin{aligned}
 V &= \pi \int_0^1 (x - x^3)^2 dx \\
 &= \pi \int_0^1 x^2 - 2x^4 + x^6 dx \\
 &= \pi \left[\frac{x^3}{3} - \frac{2x^5}{5} + \frac{x^7}{7} \right]_0^1 \\
 &= \pi \left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7} \right) \\
 &= \frac{8\pi}{105}
 \end{aligned}$$

Question 13 (e) (i)

Criteria	Marks
Provides correct sketch	1

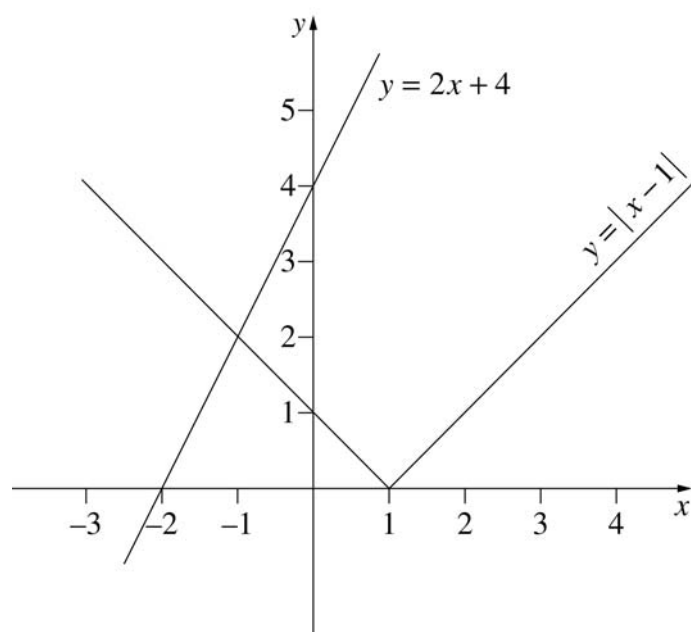
Sample answer:



Question 13 (e) (ii)

Criteria	Marks
• Provides correct solution	2
• Sketches $y = 2x + 4$ or attempts to solve $2x + 4 = -(x - 1)$, or equivalent merit	1

Sample answer:



$$2x + 4 = -(x - 1)$$

$$2x + 4 = -x + 1$$

$$3x = -3$$

$$x = -1$$

Question 14 (a)

Criteria	Marks
• Provides correct solution	2
• Attempts to integrate, or equivalent merit	1

Sample answer:

$$a = e^{2t} - 4 \qquad t = 0$$

$$v = 0$$

$$v = \int e^{2t} - 4 \, dt$$

$$v = \frac{e^{2t}}{2} - 4t + c$$

$$0 = \frac{e^{2(0)}}{2} - 4(0) + c$$

$$0 = \frac{1}{2} + c$$

$$c = -\frac{1}{2}$$

$$v = \frac{e^{2t}}{2} - 4t - \frac{1}{2}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Finds both x-values or determines the nature of one stationary point, or equivalent merit	1

Sample answer:

$$f'(x) = 0$$

$$0 = 3x^2 + 2x - 1$$

$$0 = (3x - 1)(x + 1)$$

$$x = -1$$

$$x = \frac{1}{3}$$

x	-2	-1	0
$f'(x)$	+7	0	-1

Maximum at $x = -1$

x	0	$\frac{1}{3}$	1
$f'(x)$	-1	0	+4

Minimum at $x = \frac{1}{3}$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds correct primitive, or equivalent merit	1

Sample answer:

$$f(x) = \frac{3x^3}{3} + \frac{2x^2}{2} - x + c$$

at the point (0, 4)

$$4 = 0^3 + 0^2 - 0 + c$$

$$4 = c$$

$$\therefore f(x) = x^3 + x^2 - x + 4$$

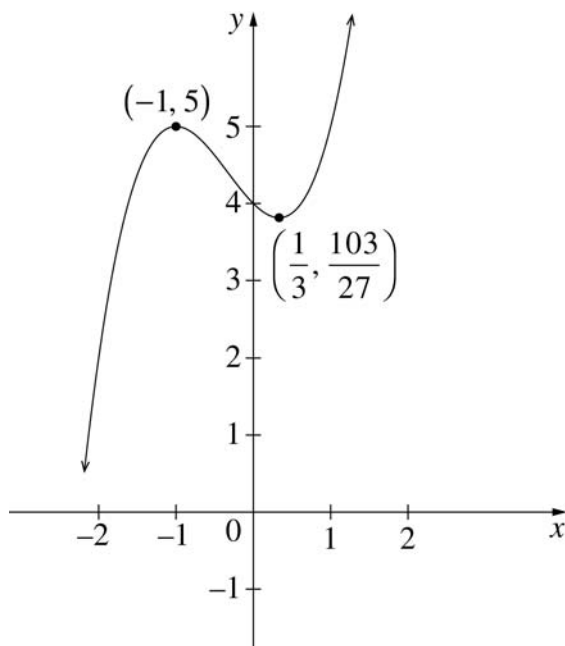
Question 14 (b) (iii)

Criteria	Marks
• Provides correct sketch	2
• Provides curve with correct shape, or equivalent merit	1

Sample answer:

$$x = -1 \qquad x = \frac{1}{3}$$

$$y = 5 \qquad y = \frac{103}{27}$$



Question 14 (b) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$f''(x) = 6x + 2$$

$$f''(x) < 0$$

$$6x + 2 < 0$$

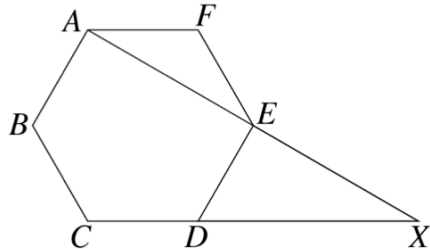
$$6x < -2$$

$$x < \frac{-1}{3}$$

Question 14 (c)

Criteria	Marks
• Provides correct solution	3
• Correctly finds two angles in $\triangle DEX$, or equivalent merit	2
• Finds vertex angle in a regular hexagon, or equivalent merit	1

Sample answer:



$ABCDEF$ is a regular hexagon of side length 1.

$$S = (6 - 2) \times 180$$

$$S = 720$$

$$\angle AFE = \frac{720^\circ}{6}$$

$$= 120^\circ$$

$\triangle AEF$ is isosceles, sides of a regular hexagon are equal.

$$\angle AEF = \frac{180^\circ - 120^\circ}{2}$$

$$= 30^\circ$$

$$\angle AED = 120^\circ - 30^\circ$$

$$= 90^\circ$$

$$\angle DEX = 90^\circ \quad (\text{supplementary } \angle)$$

$$\angle EDX = 60^\circ \quad (\text{supplementary } \angle)$$

$$ED = 1 \quad (\text{given})$$

$$\tan 60^\circ = \frac{EX}{1} \quad (\triangle DEX \text{ is right angled})$$

$$\therefore EX = \sqrt{3}$$

Question 14 (d)

Criteria	Marks
• Provides correct solution	3
• Obtains two equations in a and b , or equivalent merit	2
• Obtains the slope of the curve where $x = 2$ or shows point of tangency is $(2, -2)$, or equivalent merit	1

Sample answer:

$$y = x^3 + ax^2 + bx + 4$$

$$y' = 3x^2 + 2ax + b$$

$$y = x - 4$$

$$\therefore m = 1 \quad \text{at } x = 2$$

$$1 = 3(2)^2 + 2a(2) + b$$

$$1 = 12 + 4a + b$$

$$-11 = 4a + b \quad \text{①}$$

$$\text{at } x = 2$$

$$y = 2 - 4$$

$$y = -2$$

$$-2 = 2^3 + a(2)^2 + b(2) + 4$$

$$-2 = 8 + 4a + 2b + 4$$

$$-14 = 4a + 2b \quad \text{②}$$

$$\text{②} - \text{①}: b = -3$$

$$-11 = 4a - 3$$

$$-8 = 4a$$

$$\therefore a = -2$$

Question 15 (a)

Criteria	Marks
• Provides correct solution	2
• Obtains the quadratic equation in x , or equivalent merit	1

Sample answer:

$$e^{2\ln x} = x + 6 \quad x > 0$$

$$e^{\ln x^2} = x + 6$$

$$x^2 = x + 6$$

$$x^2 - x - 6 = 0$$

$$(x - 3)(x + 2) = 0$$

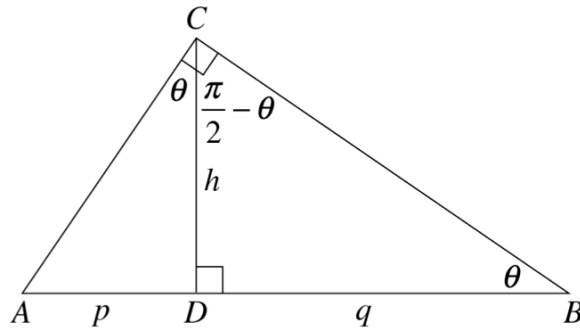
$$x = 3 \quad (\text{as } x > 0)$$

Question 15 (b)

Criteria	Marks
• Provides correct solution	3
• Show one pair of triangles are similar, or equivalent merit	2
• Attempts to show that one pair of triangles are similar or uses p or q in a Pythagorean identity, or equivalent merit	1

Sample answer:

Method 1:



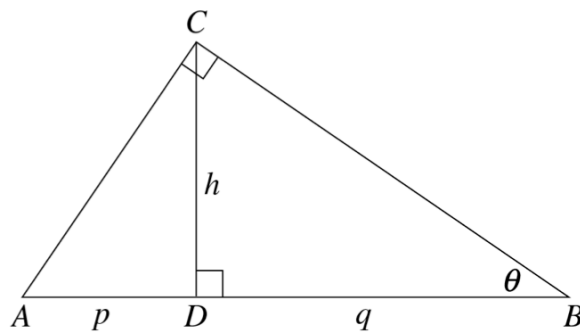
$\triangle ADC \parallel \triangle CDB$ (AA)

$$\frac{h}{p} = \frac{q}{h} \quad (\text{corresponding sides in proportion})$$

$$\therefore h^2 = pq$$

$$h = \sqrt{pq} \quad (h > 0)$$

Method 2:



Pythag in $\triangle ABC$

$$AB^2 = AC^2 + BC^2$$

$$(p + q)^2 = AC^2 + BC^2$$

Pythag in $\triangle ADC$ and $\triangle BDC$

$$(p+q)^2 = p^2 + h^2 + q^2 + h^2$$

$$p^2 + 2pq + q^2 = p^2 + 2h^2 + q^2$$

$$2pq = 2h^2$$

$$h^2 = pq$$

$$h = \sqrt{pq} \quad (h > 0)$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Shows that the length of BR is $\frac{8}{x}$, or equivalent merit	2
• Shows $\angle PBR$ is equal to $\angle APQ$, or equivalent merit	1

Sample answer:

$BR \parallel PQ$ (alternate angles).

So $\angle RBP = \angle QPA$ (corresponding angles).

$\angle BRP$ and $\angle PQA$ are both right angles.

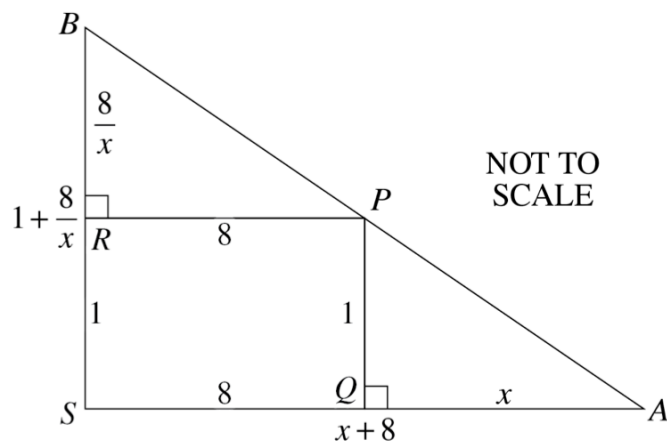
So $\triangle BRP$ and $\triangle PQA$ are similar (AA).

$$\frac{BR}{RP} = \frac{PQ}{QA}$$

$$\frac{BR}{8} = \frac{1}{x}$$

$$BR = \frac{8}{x}$$

BR and QA produced will meet at right angles at S .



$$\text{So } D^2 = (x+8)^2 + \left(\frac{8}{x} + 1\right)^2$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Verifies that a stationary point occurs at $x = 2$, or equivalent merit	2
• Differentiates D^2 , or equivalent merit	1

Sample answer:

$$\frac{d(D^2)}{dx} = 2(x+8) + 2\left(\frac{8}{x} + 1\right)\left(\frac{-8}{x^2}\right)$$

$$\begin{aligned} \text{at } x = 2 \quad \frac{d(D^2)}{dx} &= 2 \times 10 + 2 \times 5 \times (-2) \\ &= 20 - 20 \\ &= 0 \end{aligned}$$

So has a stationary point at $x = 2$

$$\begin{aligned} \text{at } x = 1 \quad \frac{d(D^2)}{dx} &= 2 \times 9 + 2 \times 9 \times (-8) \\ &= 18 - 144 \\ &< 0 \end{aligned}$$

$$\begin{aligned} \text{at } x = 4 \quad \frac{d(D^2)}{dx} &= 2 \times 12 + 2 \times 3 \left(\frac{-1}{2}\right) \\ &= 24 - 3 \\ &> 0 \end{aligned}$$

Summary

x	1	2	4
$\frac{d(D^2)}{dx}$	$\searrow$	$-$	$\nearrow$

So D^2 has a minimum at $x = 2$.

Question 15 (d) (i)

Criteria	Marks
• Provides correct solution	2
• Considers a complementary event, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 P(\text{at least 1}) &= 1 - P(\text{none}) \\
 &= 1 - (1 - 0.02)^2 \\
 &= 0.0396
 \end{aligned}$$

Question 15 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Obtains an inequality or equation for the number of people, or equivalent merit	1

Sample answer:For n people

$$\begin{aligned}
 P(\text{at least 1}) &= 1 - P(\text{none}) \\
 &= 1 - (1 - 0.02)^n \\
 &= 1 - 0.98^n
 \end{aligned}$$

$$1 - 0.98^n \geq 0.4$$

$$0.6 \geq 0.98^n$$

$$\ln 0.6 \geq \ln(0.98^n) = n \ln 0.98$$

$$n \geq \frac{\ln 0.6}{\ln 0.98} = 25.28\dots$$

So $n = 26$ is smallest number.

Question 16 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains the expression for A_1 , or equivalent merit	1

Sample answer:

$$\text{Quarterly rate} = \frac{6}{100} \div 4$$

$$A_1 = 1\,000\,000(1.015)^4 - 80\,000$$

$$A_2 = A_1(1.015)^4 - 80\,000$$

$$= 1\,000\,000(1.015)^8 - 80\,000(1.015)^4 - 80\,000$$

$$= 1\,000\,000(1.015)^8 - 80\,000(1 + 1.015^4)$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	3
• Correctly sums the series and equates A_n to zero, or equivalent merit	2
• Obtains an expression for A_n , or equivalent merit	1

Sample answer:

Continuing in this way,

$$\begin{aligned}
 A_n &= 1\,000\,000(1.015)^{4n} - 80\,000(1 + 1.015^4 + 1.015^8 + \dots + 1.015^{4(n-1)}) \\
 &= 1\,000\,000(1.015)^{4n} - 80\,000 \times \frac{(1.015)^{4n} - 1}{(1.015^4) - 1}
 \end{aligned}$$

(geometric series with $a = 1$, $r = 1.015^4$, n terms).

Let $K = 1.015^{4n}$

We want to find n so that $A_n = 0$.

$$\text{ie } 1\,000\,000K = \frac{80\,000(K - 1)}{0.06136}$$

$$0.06136 \times 100K = 8K - 8$$

$$6.136K = 8K - 8$$

$$\therefore 1.864K = 8$$

$$\therefore (1.015)^{4n} = \frac{8}{1.864} = 4.2918\dots$$

$$4n \log(1.015) = \log(4.2918\dots)$$

$$n = \frac{1 \log(4.2918)}{4 \log(1.015)}$$

$$\approx 24.46$$

$\therefore$ Money can be withdrawn for 24 years.

Question 16 (b)

Criteria	Marks
• Provides correct solution	3
• Uses $t = 0$, $\frac{\pi}{4}$ and $\frac{\pi}{2}$ in Simpson's rule, or equivalent merit	2
• Finds the time at which the particle first comes to rest, or equivalent merit	1

Sample answer:

Particle first comes to rest when $v = 0$.

$$\therefore e^{\cos t} = 1 \quad \therefore \cos t = 0$$

$$\therefore t = \frac{\pi}{2}$$

Distance travelled corresponds to area of region under the curve between $t = 0$ and $t = \frac{\pi}{2}$

$$\text{distance} = \int_0^{\frac{\pi}{2}} (e^{\cos t} - 1) dt$$

Using Simpson's Rule:

$$\frac{1}{6} \left(\frac{\pi}{2} - 0 \right) \left[\left(e^{\cos 0} - 1 \right) + 4 \left(e^{\cos \frac{\pi}{4}} - 1 \right) + \left(e^{\cos \frac{\pi}{2}} - 1 \right) \right]$$

$$= \frac{\pi}{12} \left[e - 1 + 4e^{\frac{1}{\sqrt{2}}} - 4 + 0 \right]$$

$$= \frac{\pi}{12} \left[e + 4e^{\frac{1}{\sqrt{2}}} - 5 \right]$$

$$\doteq 1.53 \text{ (to 2 decimal places)}$$

Question 16 (c) (i)

Criteria	Marks
Provides correct solution	2
Finds the equation of the tangent at (1, 1) or finds the equation of the line through (1, 1) and $\left(\frac{r-1}{r}, 0\right)$, or equivalent merit	1

Sample answer:

$$y = x^r$$

$$\frac{dy}{dx} = rx^{r-1}$$

when $x = 1$ $\frac{dy}{dx} = r$

Equation of tangent is $y - 1 = r(x - 1)$.

when $y = 0$ $-1 = r(x - 1)$

$$\therefore x - 1 = \frac{-1}{r}$$

$$\therefore x = 1 - \frac{1}{r} = \frac{r-1}{r}$$

Hence, the line meets the x -axis at $\left(\frac{r-1}{r}, 0\right)$

Question 16 (c) (ii)

Criteria	Marks
Provides correct solution	2
Recognises that the area of R is the difference of two areas, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{Area of } R &= \int_0^1 x^r dx - \text{area of triangle} \\
 &= \left[\frac{x^{r+1}}{r+1} \right]_0^1 - \frac{1}{2} \times \left(1 - \frac{r-1}{r} \right) \times 1 \\
 &= \frac{1}{r+1} - \frac{1}{2} \left(\frac{r - (r-1)}{r} \right) \\
 &= \frac{1}{r+1} - \frac{1}{2r} \\
 &= \frac{2r - (r+1)}{2r(r+1)} = \frac{r-1}{2r(r+1)}
 \end{aligned}$$

Question 16 (c) (iii)

Criteria	Marks
• Provides correct solution	3
• Finds the value of r at which there is a stationary point, or equivalent merit	2
• Finds the derivative of the area of R , or equivalent merit	1

Sample answer:

$$\text{Let } A = \frac{r-1}{2r(r+1)}$$

$$\frac{dA}{dr} = \frac{2r(r+1) \cdot 1 - (r-1)(4r+2)}{4r^2(r+1)^2}$$

$$\frac{dA}{dr} = 0 \text{ when } 2r(r+1) - (r-1)(4r+2) = 0$$

$$-2r^2 + 4r + 2 = 0$$

$$r^2 - 2r - 1 = 0$$

$$r = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$$

Since $r \geq 0$ we have $r = 1 + \sqrt{2}$

$$\text{When } r = 2 \quad \frac{dA}{dr} = \frac{2}{\text{positive}} > 0$$

$$\text{When } r = 3 \quad \frac{dA}{dr} = \frac{-4}{\text{positive}} < 0$$

$\therefore$ maximum value of A occurs when $r = 1 + \sqrt{2}$.

2019 HSC Mathematics Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.1	P3
2	1	1.4	P3
3	1	1.1	P3
4	1	9.5	P5
5	1	12.3	H3
6	1	3.2	H5
7	1	13.3	H5
8	1	14.3	H5
9	1	13.6	H5
10	1	14.3	H5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	5.5	P3
11 (b)	2	8.8,13.5	H5
11 (c)	2	8.9	P7
11 (d)	2	7.3	H5
11 (e)	3	11.2	H5
11 (f)	2	3.2	H5
11 (g)	2	4.2,4.4	P4
12 (a) (i)	2	6.2	P4
12 (a) (ii)	2	6.8	P4
12 (b)	3	7.1	H5
12 (c) (i)	1	14.2	H3, H4
12 (c) (ii)	2	14.2	H3, H4
12 (c) (iii)	2	14.2	H3, H4, H5
12 (d)	3	11.4,12.5	H8
13 (a)	3	1.4,5.3,13.3	H5
13 (b)	3	5.5,13.1	H5
13 (c) (i)	2	12.4	H5
13 (c) (ii)	1	12.5,11.1	H5
13 (d)	3	11.4	H8
13 (e) (i)	1	1.2,4.2	P4
13 (e) (ii)	2	1.2,1.4,4.2	P4
14 (a)	2	12.5,14.1	H5
14 (b) (i)	2	10.2	H6

Question	Marks	Content	Syllabus outcomes
14 (b) (ii)	2	10.8	H6
14 (b) (iii)	2	10.5	H6
14 (b) (iv)	1	10.4	H6
14 (c)	3	6.8,5.3	H5
14 (d)	3	8.7,10.7	H5
15 (a)	2	12.2,1.4	H3
15 (b)	3	2.5	H5
15 (c) (i)	3	2.5	H4
15 (c) (ii)	3	10.6	H5
15 (d) (i)	2	3.2,3.3	H5
15 (d) (ii)	2	3.2,3.3,12.2	H3, H5
16 (a) (i)	2	7.5	H5
16 (a) (ii)	3	7.5	H5
16 (b)	3	11.3,14.3	H3, H5, H8
16 (c) (i)	2	10.7	H5
16 (c) (ii)	2	11.4	H5, H8
16 (c) (iii)	3	10.6	H5